

Wildfire Fuel Risk Scoring

A Terrain-Based Peril Assessment Prototype

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Data source: sklearn.fetch_covtype (Roosevelt National Forest, CO)

Executive Summary

This project builds a prototype quantitative risk-scoring model for physical site risk, using publicly available forest cover and terrain data from Roosevelt National Forest, Colorado, as a proof-of-concept dataset. Each record is assigned a continuous risk score, on a scale of 1 to 10, that combines domain-informed fuel characteristics with terrain factors known to influence wildfire behavior, namely slope, solar exposure, and proximity to historical fire activity. A Random Forest regression model is then trained to estimate this composite score from terrain and soil characteristics alone, without knowing the underlying species that is actually present. The model achieves an R^2 of 0.86 on held-out data, although this result is examined more closely later in the report, since part of it reflects terrain inputs that directly inform how the score was constructed in the first place. The more meaningful signal is really the model's ability to infer species-driven risk using only indirect proxies. Taken together, this forms a full quantitative risk-assessment pipeline, going from domain-driven feature construction, through predictive modeling, validation, and residual analysis, that can generalize to physical-risk assessment contexts beyond wildfire, including site selection and infrastructure risk evaluation.

Methodological Parallel

This approach mirrors a common quantitative risk-assessment pipeline. Domain knowledge is first encoded into structured, quantifiable inputs, and a statistical model is then trained and validated against that structure in order to produce a predictive, generalizable output. In this case, qualitative fire-ecology knowledge, meaning which species and terrain conditions actually carry more risk, is converted into a quantitative score, and a Random Forest model is validated based on its ability to recover that score using only indirect terrain and soil data, without being told what is growing there in the first place. This same structure, taking qualitative domain research and translating it into a quantitative, validated model, applies broadly to physical risk and site-viability assessment as a whole, including infrastructure and facility siting decisions.

Methodology

The dataset covers 581,012 forest observations across seven different cover types, each with terrain attributes including elevation, slope, aspect, hillshade, and distance to hydrology, roadways, and historical fire points. A baseline fuel score, on the same 1 to 10 scale, was assigned to each species based on known fire-ecology characteristics. For example, resinous conifers such as Douglas-fir and Lodgepole Pine score the highest, while moisture-retaining deciduous species like Aspen score the lowest. Krummholz, which is a stunted, high-elevation conifer growth form, was scored as a moderate risk in order to reflect its sparse fuel continuity, despite technically still being a conifer.

This baseline was then adjusted using three terrain factors that are known to affect real fire behavior: solar exposure, since south and southwest-facing slopes tend to dry out faster, slope steepness, since fire spreads faster going uphill, and proximity to historical fire activity. Once combined, the score was rescaled back to a 1 to 10 range. A Random Forest regressor was then trained to predict this composite

score using only terrain and soil features, with species identity deliberately excluded, in order to test whether risk could be reasonably estimated in areas where species-level data isn't available in the first place.

Model Validation

The model achieved an R^2 of 0.859 and a mean absolute error of 0.14, on the 1 to 10 scale, using held-out test data. Because part of the risk score is a direct function of terrain inputs that the model can see, this level of accuracy is somewhat expected. The more meaningful result is that the model is also able to infer the underlying species-driven fuel component using only indirect proxies, namely elevation, soil type, and hillshade, all without ever being told what is actually growing there.

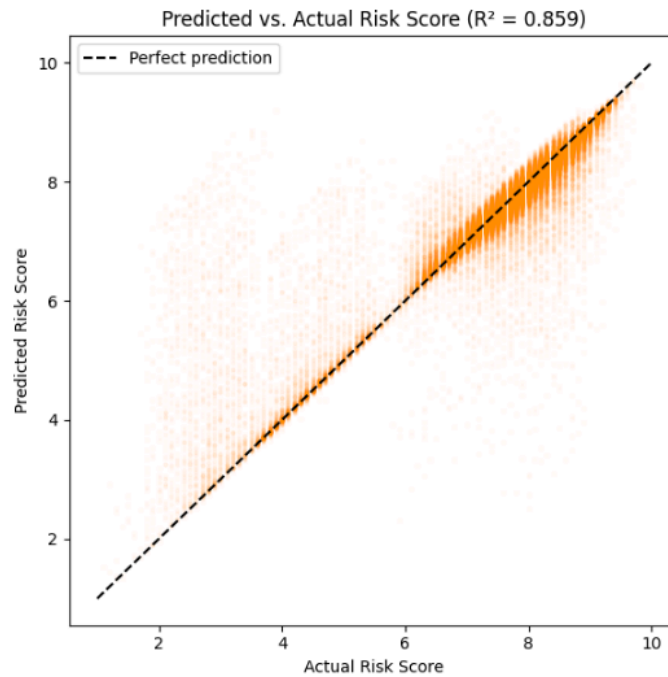


Figure 1. Predicted vs. actual risk score on held-out test data.

Key Findings

Elevation is by far the strongest predictor of risk score, which lines up with how tree species distribute by elevation zone throughout the Rockies. Proximity to historical fire points and roadways, both of which are realistic risk indicators, rank next, followed by hillshade and hydrology-related features, which end up capturing sun exposure and moisture effects in a more indirect way. It's worth noting that the top six features account for roughly 70% of the model's total decision-making, meaning risk can be reasonably approximated using only a small set of easily obtainable terrain measurements, namely elevation, fire history, and road proximity, without requiring a full soil survey to be conducted. This has a practical implication that goes beyond this dataset alone: a lightweight version of this model could plausibly run using terrain data by itself, in situations where detailed ground-truth surveys simply aren't available yet.

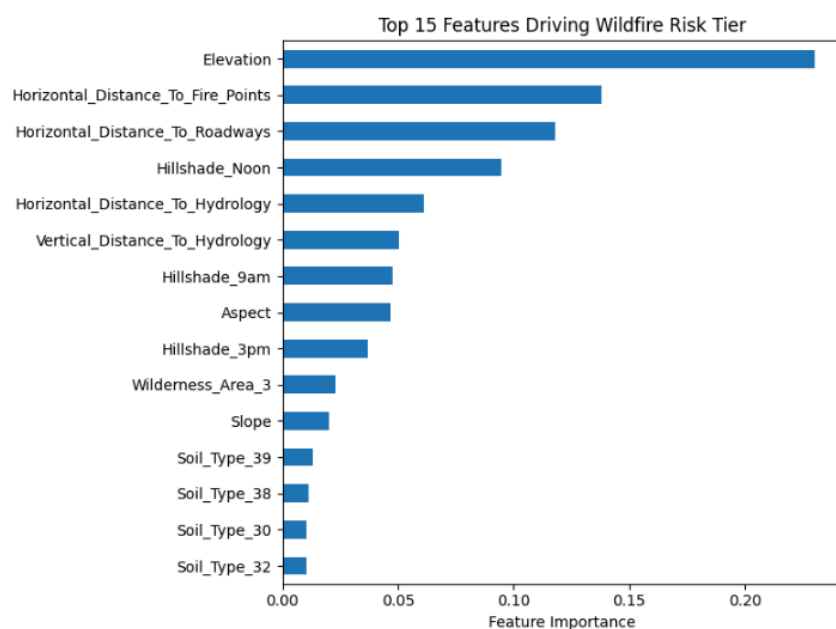


Figure 2. Top 15 features driving the risk score, by importance.

Geographic Risk Pattern

Plotting the mean risk score across elevation and aspect bins shows a clear, ecologically consistent pattern. Risk peaks at mid-elevation, roughly between 2,850 and 3,250 meters, on south and southwest-facing terrain, and it tapers off at both the lower elevations, where moister, less resinous species are more common, and at the highest elevations, where Krummholz cover becomes sparser.

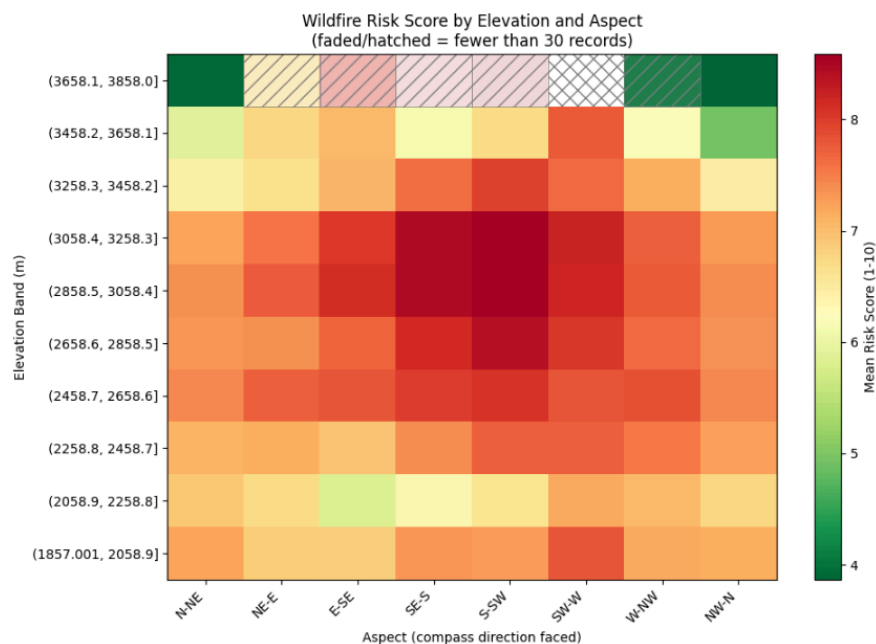


Figure 3. Mean wildfire risk score by elevation band and aspect, with cell opacity and hatching indicating sample size (faded/hatched cells have fewer than 30 records; solid white indicates no data).

Note: the highest elevation band, 3,658 to 3,858 meters, is sparsely sampled, as shown by the faded and hatched cells above, so its risk estimates should be treated with lower confidence than the well-populated mid-elevation rows.

Residual Analysis

In order to validate the model's quantitative behavior beyond just a single R^2 figure, prediction residuals, meaning the actual risk score minus the predicted one, were examined across the entire range of risk scores. The mean residual came out to be effectively zero, at -0.001, with a standard deviation of 0.41, which confirms there isn't an overall directional bias. In other words, the model does not systematically over-predict or under-predict on average.

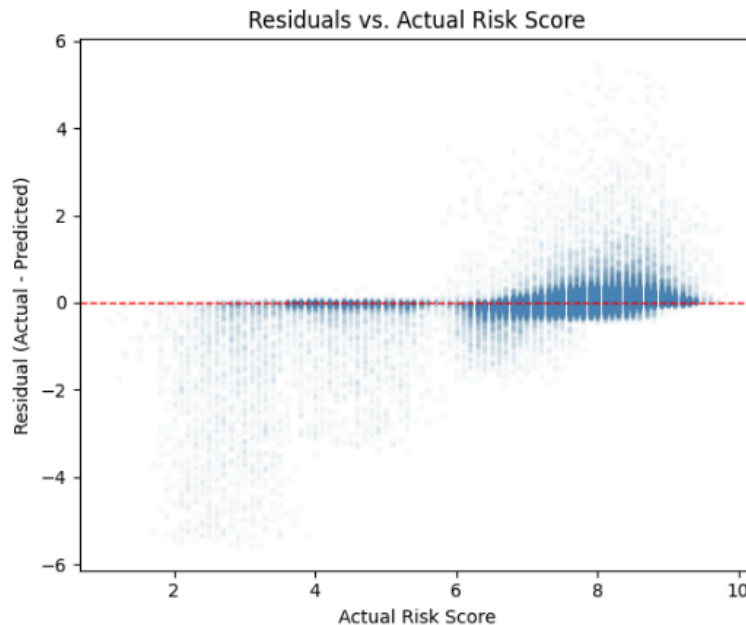


Figure 4. Prediction residuals vs. actual risk score.

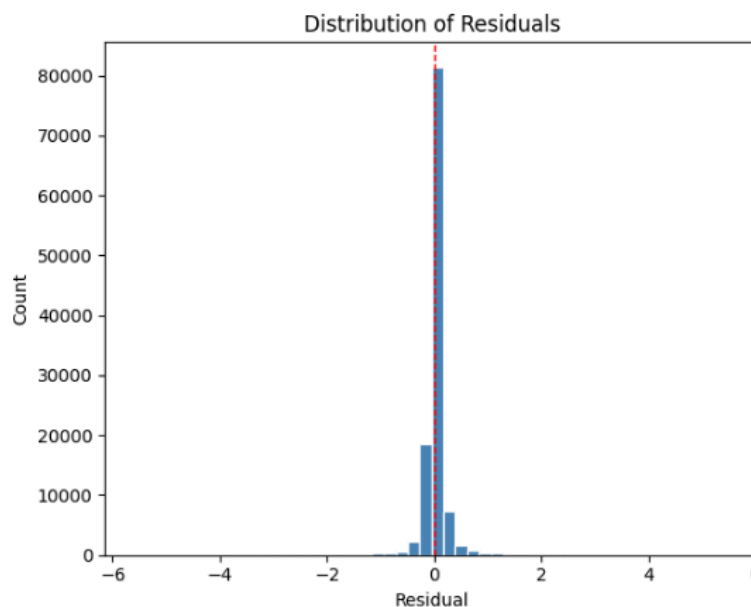


Figure 5. Distribution of prediction residuals.

Residual variance widens noticeably for actual risk scores above roughly 6, while the low-to-mid scores show tight, near-zero residuals. This pattern is really a reflection of a feature-design choice rather than a modeling deficiency. Several species with overlapping terrain profiles, namely Ponderosa Pine,

Spruce/Fir, Lodgepole Pine, and Douglas-fir, carry base fuel scores somewhere between 6.5 and 9.0, and because species identity was deliberately excluded from the model's inputs, terrain data alone cannot always distinguish between them in places where their habitats overlap. Addressing this would call for explicitly modeling prediction uncertainty, such as through quantile regression, or introducing more granular terrain and soil interaction features, rather than simply increasing model complexity. The ambiguity here is inherent to the terrain-only design that was chosen, not a capacity limitation of the Random Forest itself.

Species Risk Profile

The baseline fuel scores below reflect known fire-ecology differences between species, independent of terrain:

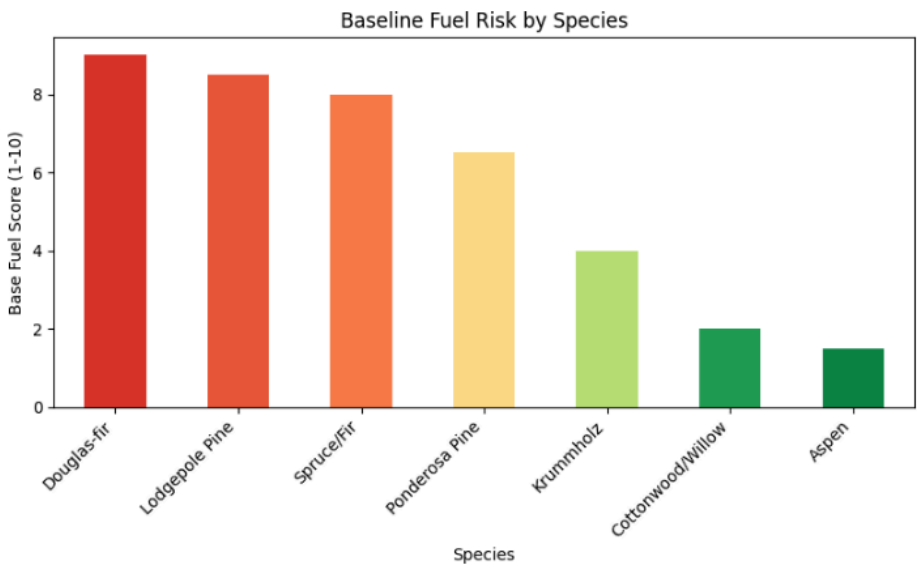


Figure 6. Baseline fuel risk score by species.

Mapping the Score to a Decision

A continuous risk score really only has practical value if it can be translated into some kind of categorical action. As a conceptual illustration of that mapping, and not meant as a calibrated recommendation, the 1 to 10 score can be split into ordinal bands, each one suggesting a different level of attention:

Risk Category	Score Range	Suggested Interpretation
Low	1.0–3.0	Minimal terrain-driven concern
Moderate	3.1–6.0	Standard monitoring
Elevated	6.1–8.5	Warrants closer review
Severe	8.6–10.0	Recommend mitigation before proceeding

This is only one of several ways a continuous score could be operationalized. A production version of this would replace these illustrative bands with thresholds that are actually calibrated against real outcome data.

Land Development Use Case

The most direct application of this scoring approach is in site selection for land development, especially for developers who are considering clearing forested parcels to build new complexes or subdivisions. Instead of treating wildfire exposure as a simple yes or no judgment, having a continuous risk score means multiple candidate sites can be compared side by side on the same scale, even if they differ quite a bit in tree species, elevation, and terrain.

Why does this matter for a developer?: Because the score doesn't just tell you a site is risky, it gives a starting point for estimating how much wildfire mitigation that site would actually require. This could include defensible space clearing, fire-resistant landscaping buffers, or how access roads should be planned, all of which need to be accounted for before a purchase or clearing decision is finalized. The elevation and aspect grid built earlier in this report adds another layer of usefulness here as well, since it can be used within a single parcel, not just across separate ones. This means it could help identify which portions of a site face the least wildfire exposure, which would be valuable information when deciding where to place structures or access points. When used this way, wildfire exposure moves from being a late, expensive surprise, to an upfront and quantifiable part of the site acquisition and design process.

Limitations & Next Steps

- Fuel scores and terrain weights are expert-judgment estimates, and are not calibrated against historical outcome data.
- The dataset doesn't include real latitude or longitude, so the geographic pattern shown here is a relative elevation and aspect grid, not an actual geocoded map.
- The highest-elevation band has a limited sample size, so it should be treated as lower-confidence.
- The illustrative score bands are conceptual, meant to demonstrate the scoring-to-decision concept rather than serve as calibrated thresholds.
- A production version would benefit from real site-level geospatial data, historical outcome records for calibration, and a proper credibility-weighting approach.